



Disha Tomar

Software Engineer (Full-Stack / AI) | IIT Kharagpur

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LinkedIn · Portfolio

SUMMARY

- AI Engineer / Full Stack Developer with 4+ years of experience building scalable web applications using React, Angular, Next.js, Node.js, and FastAPI. Skilled in developing RESTful APIs and working with PostgreSQL and MongoDB to deliver efficient, data-driven solutions.
- Comfortable owning delivery end to end, from architecture and CI/CD on AWS (EC2, App Runner, CloudWatch) to clean, performant, user-centric interfaces.
- Hands-on experience leveraging AI-assisted engineering tools (Claude Code, Cursor, Claude API) daily for code generation, documentation, and accelerated delivery, building production AI-powered features rather than just using AI as a coding aid.

EXPERIENCE

Senior Software Engineer

Oct 2025 — Present

Alucor Limited · Dubai, United Arab Emirates

- Sole engineer designing, building, and deploying a production Document Management System (DMS) for large-scale construction documentation, with full ownership across frontend, backend, data, AI features, and infrastructure.
- Architected and **shipped a full-stack** DMS using **Next.js 14 (App Router, React Server Components, TypeScript) and FastAPI**, delivering structured document numbering, revision and version control, and multi-step workflow-driven lifecycle management that replaced error-prone Excel processes.
- Built the entire frontend as a feature-modular React architecture with React Query for server state, react-hook-form and Zod for type-safe forms, and a shadcn/ui and Tailwind design system, delivering responsive, accessible (ARIA/WCAG) interfaces for document upload, registers, transmittals, and dashboards.
- Designed a layered FastAPI backend (router, service, CRUD, SQLAlchemy) over PostgreSQL, with Alembic migrations, strong referential-integrity constraints, and a pluggable storage abstraction (Local, S3, Azure Blob, Alfresco ECM) supporting versioned files, metadata indexing, and full audit logging.
- Engineered an **AI-powered document intelligence pipeline**: background workers that convert uploaded PDFs and Word docs to Markdown, fall back to local OCR (RapidOCR and ONNX) for scanned or image-only files with zero cloud cost, and generate concise document summaries using the Anthropic Claude API, making thousands of legacy documents searchable and reviewable.
- Implemented enterprise authentication and authorization with Keycloak, designing RBAC and ABAC policies for fine-grained, context-aware access control across organizations, projects, and workflows.
- Containerized a multi-service stack (FastAPI, PostgreSQL, Keycloak, background workers) with Docker and Docker Compose, and built CI/CD pipelines and AWS deployments (EC2, App Runner) with CloudWatch monitoring for proactive health and performance tracking.
- Modeled a complex domain end-to-end, covering structured document-number generation, revision lifecycle state machines, and inter-organization transmittal handoffs, translating real construction-industry workflows into a maintainable schema and API.

Software Engineer II

Jun 2023 — Sep 2025

Korn Ferry · Remote - India

- I worked for Korn Ferry Digital, the product and technology division of Korn Ferry, focused on delivering data-driven talent solutions and scalable assessment platforms for global enterprises.
- Led the frontend development for micro frontend based assessment platform, working on both React and Angular (16+) applications.
- Integrated an AI-powered proctoring service into the assessment platform's frontend, built in React and Typescript, enabling seamless exam monitoring, enhancing test integrity, resulting in a 30% reduction in unauthorized activities.
- Optimized modular frontend architecture (Angular + TypeScript) project setup to enable live code updates from shared libraries using Webpack and path mapping, reducing rebuild time by 94% and improving developer workflow.

- Enhanced platform accessibility in React application through semantic HTML and ARIA standards, resulting in a 25% improvement in Lighthouse accessibility scores and ensuring compliance with WCAG guidelines, thereby delivering secure, scalable digital products and improving user experience for people with disabilities.
- Extensively worked with Handlebars (HBS) templates and collaborate with the data science team to build complex, math-driven graph components by transforming statistical models into intuitive and interactive visualizations.
- Developed and maintained comprehensive unit tests using React Testing Library, Jest, ensuring 80%+ code coverage, which significantly improved code reliability and reduced production bugs.
- Optimized AWS Lambda performance by upgrading Node.js, reducing bundle size by 30% and improving load time by 25%.
- Track and analyze logs in AWS CloudWatch to debug and enhance application performance, proactively identifying issues, reducing downtime and ensuring system reliability.
- Conduct code reviews and drive technical design for UI feature implementation. Participate in technical interviews.

Senior Software Developer

Unvoid Tech Studio Pvt Ltd. · Remote - India

Mar 2023 — May 2023

- Led frontend development for an accounting platform managing business finances, using React, Redux, Tailwind and Material UI; partnered closely with the backend team on REST API integration for smooth data flow and a consistent UX.
- Built and deployed the company's marketing website with Next.js and GSAP.
- Mentored the frontend team on modern frameworks, code quality and best practices.

Frontend Developer

Wiztric Technologies Private Ltd. · Bengaluru - India

May 2022 — Feb 2023

- Single-handedly developed and deployed the MVP of the desktop-based SCADA system interface using React, Material UI, Python (Flask) HTML5, CSS3, and Electron, streamlining electrical substation automation and reducing manual intervention by 40%.
- Implemented real-time data visualization using webSocket, Ngx-charts and Plotly.js.

Assistant Town Planner

Directorate of Local Bodies · Jaipur, India

May 2017 — May 2022

- This was a non-technical role where I worked as an architect and town planner. During this time, I gained strong problem-solving skills **before transitioning into the tech industry to pursue my passion for software development.**

EDUCATION

Master's Degree

Indian Institute of Technology, Kharagpur - India

GPA: 8.18

Jul 2015 — May 2017

Bachelor's Degree

Malaviya National Institute of Technology, Jaipur - India

GPA: 8.16

Jul 2009 — May 2014

SKILLS

Frontend: Javascript (ES6+), Typescript, HTML5 and CSS3, Responsive Design, Redux, React, Angular (16+), Next.js, Material UI, Accessibility (ARIA), Handlebars (HBS) templates, D3.js, Tailwind, React Server Components, Modern CSS Architecture

Backend: Node.js, Python, FastAPI, Web security (CORS, CSP, JWT), PostgreSQL, SQLAlchemy, MongoDB , Secure Coding Practices, RBAC / ABAC

Testing: Jest, React Testing Library

CS Fundamentals: Data Structures and Algorithms, Browser Internals (event loop, rendering, memory)

Tooling & Infrastructure: Jira, NPM, Webpack, Git, Github, Kafka, AWS, Docker

AI/ML : LLM integration (Anthropic Claude API, Gemini), OCR (RapidOCR/ONNX),, Document parsing, Claude Code, Cursor

PROJECTS

AI-Powered Resume Builder (In Development)

A full-stack web app (React, TypeScript, Supabase, Google Gemini) for creating job-tailored resumes from a single master profile, featuring AI-driven job-description analysis, tailored suggestions, PDF export, etc. This resume was created using this app.